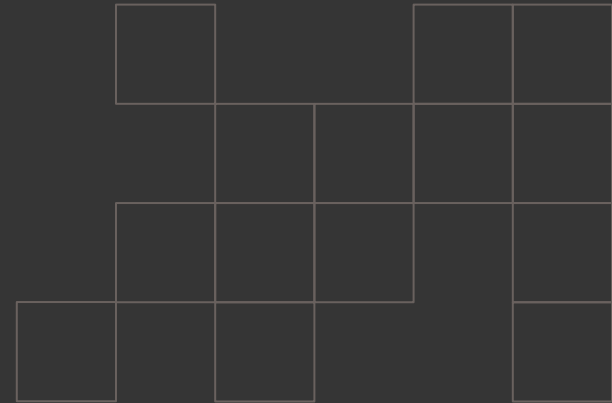


Michael Nuttall,
Colin Duckworth, Colten Ross

Evaluating Mushroom Edibility

Spring 2026

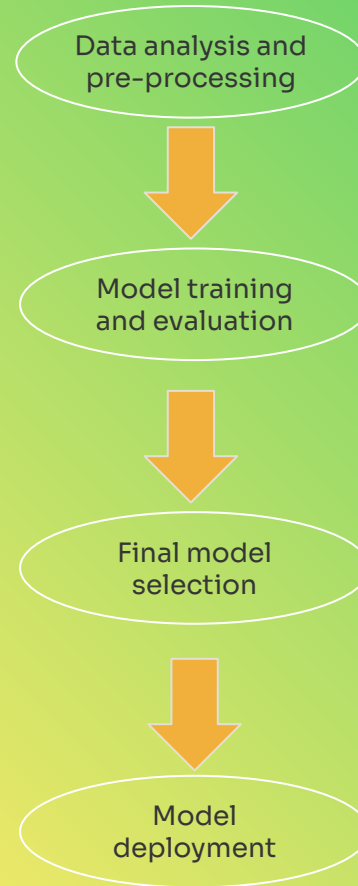


Machine learning classification algorithms on a mushroom dataset.

Dataset contains mushrooms physical descriptions as 22 features correlating to a classification of edible or poisonous.

This dataset contains samples corresponding to 23 species of gilled mushrooms drawn from The Audubon Society Field Guide to North American Mushrooms (1981).

Tasks



Data Pre-Processing

Potential challenge

How we responded

1

Missing or incomplete entries in the dataset.

We replaced missing and unknown data entries with the mean value of the column for the row index missing data.

2

Non numerical data values.

All of the data was encoded using the one hot encoder method to transfer the data from letter categories into numerical values for the model.

3

Scaling of the encoded data.

The data was purely categorical so the numerical values required no scaling to allow proper model behavior.

Model evaluation and testing

How we'll compare different models

Step 1

Create a modularized setup which breaks data processing, training, and evaluation into separate steps.

Step 2

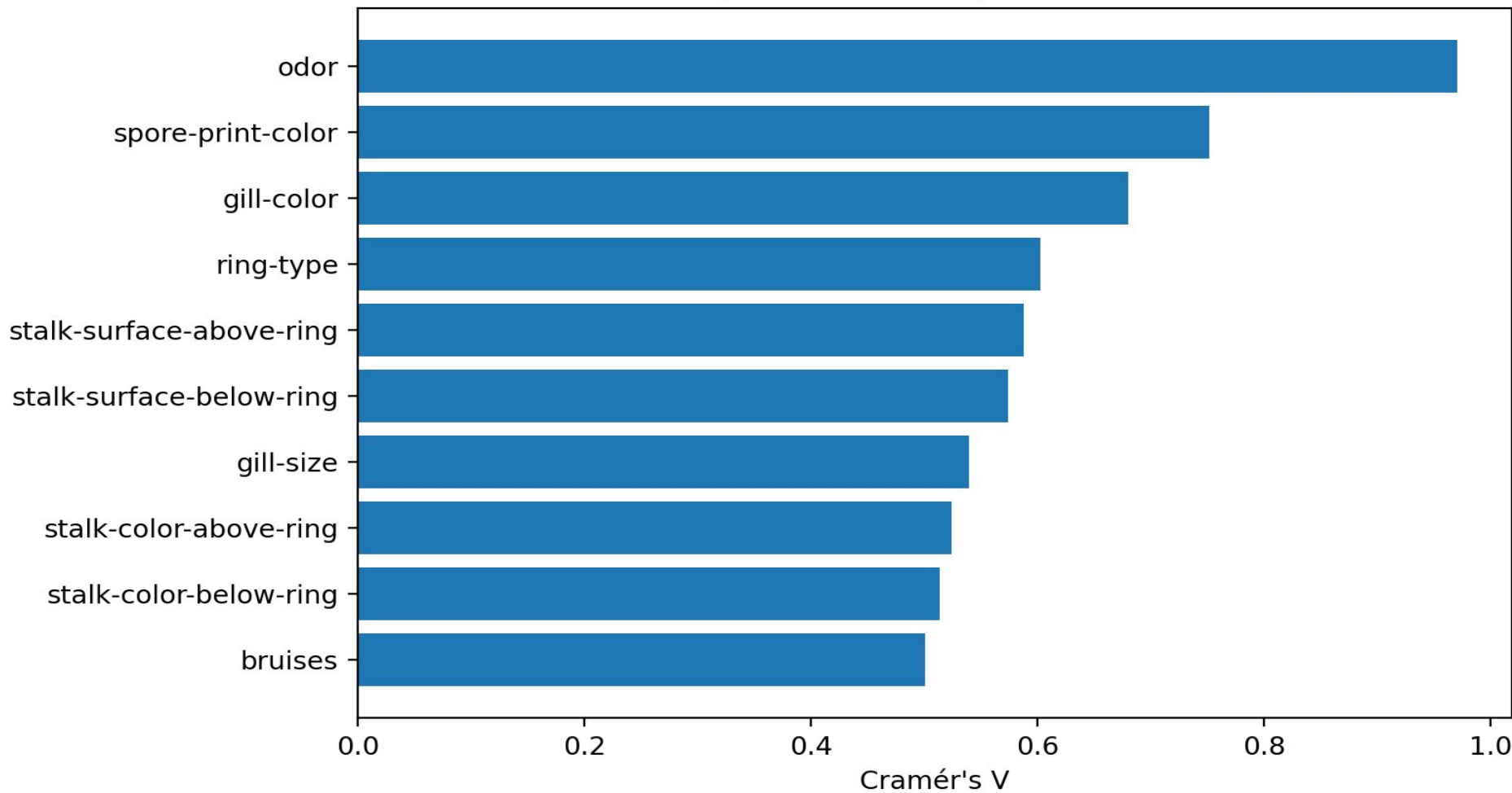
Use the pre-processed data to train various different models and compare them using the same metrics.

Step 3

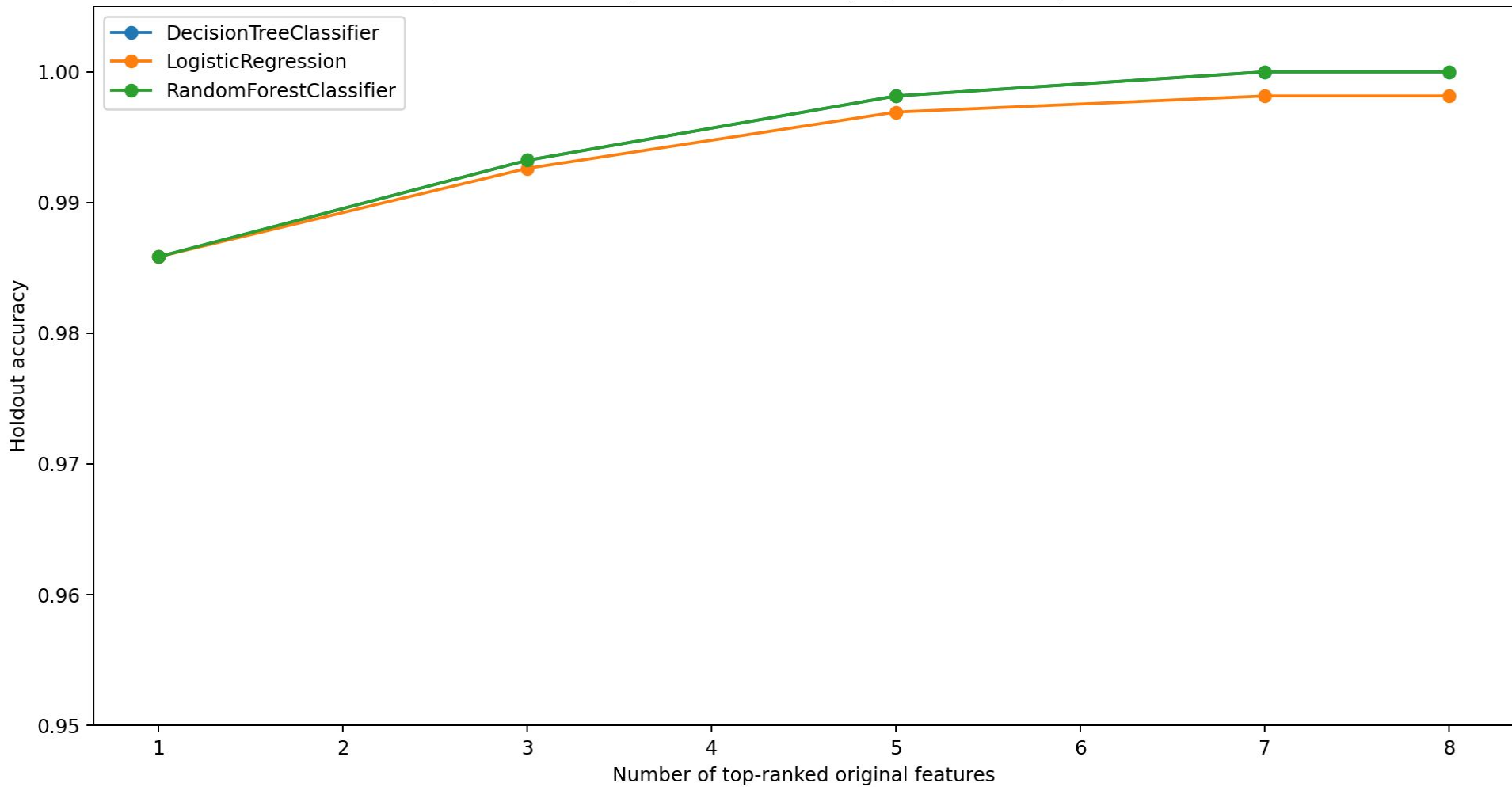
Identify the top performing/most accurate models for further evaluation.

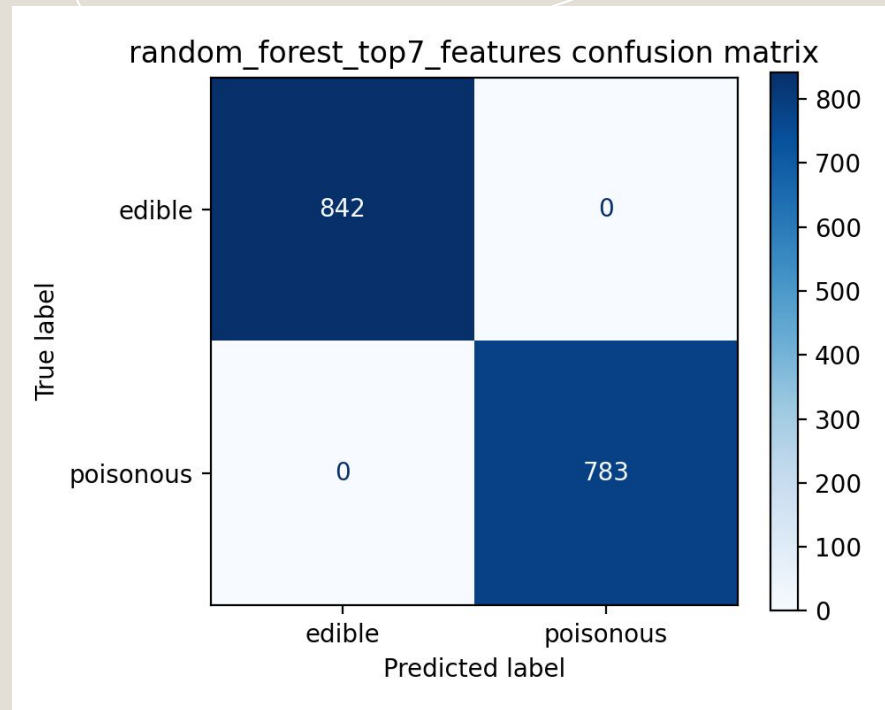
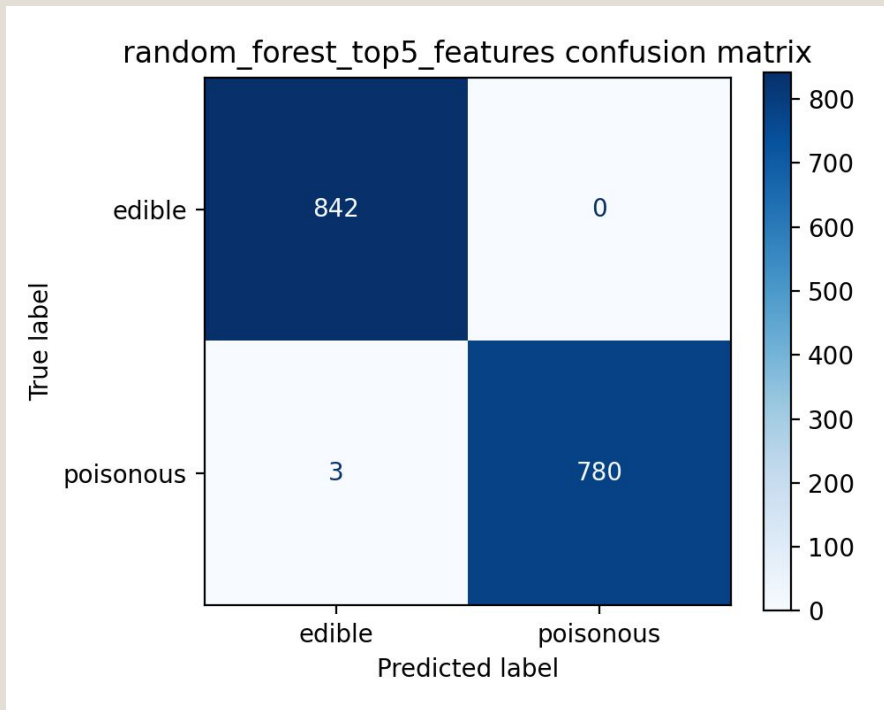
experiment_name	model_class	accuracy	f1	cv_accuracy_mean	cv_f1_mean	[8] Running decision_tree_top3_features ...
decision_tree_split_70_30_cv3	DecisionTreeClassifier	1.000000	1.000000	0.999631	0.999617	[9] Running decision_tree_top5_features ...
decision_tree_split_70_30_cv5	DecisionTreeClassifier	1.000000	1.000000	0.999631	0.999616	[10] Running decision_tree_top7_features ...
decision_tree_top7_features	DecisionTreeClassifier	1.000000	1.000000	1.000000	1.000000	[11] Running decision_tree_top8_features ...
decision_tree_top8_features	DecisionTreeClassifier	1.000000	1.000000	1.000000	1.000000	[12] Running decision_tree_topall_features ...
logreg_engineered	LogisticRegression	1.000000	1.000000	1.000000	1.000000	[13] Running random_forest_top1_features ...
logreg_engineered_comparison	LogisticRegression	1.000000	1.000000	1.000000	1.000000	[14] Running random_forest_top3_features ...
random_forest_baseline	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	[15] Running random_forest_top5_features ...
random_forest_engineered_comparison	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	[16] Running random_forest_top7_features ...
random_forest_no_engineering_comparison	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	[17] Running random_forest_top8_features ...
random_forest_split_30	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	[18] Running random_forest_topall_features ...
random_forest_split_70_30_cv3	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	[1] Running logreg_no_engineering_comparison ...
random_forest_split_70_30_cv5	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	[2] Running logreg_engineered_comparison ...
random_forest_split_80_20_cv3	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	[3] Running decision_tree_no_engineering_comparison ...
random_forest_split_80_20_cv5	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	[4] Running decision_tree_engineered_comparison ...
random_forest_top7_features	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	[5] Running random_forest_no_engineering_comparison ...
random_forest_top8_features	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	[6] Running random_forest_engineered_comparison ...
random_forest_topall_features	RandomForestClassifier	1.000000	1.000000	1.000000	1.000000	
xgboost_baseline	XGBClassifier	1.000000	1.000000	1.000000	1.000000	Top-k sweep summary:
xgboost_top8	XGBClassifier	1.000000	1.000000	1.000000	1.000000	experiment_name
logreg_split_70_30_cv3	LogisticRegression	0.998769	0.998722	0.999754	0.999744	model_class
logreg_split_70_30_cv5	LogisticRegression	0.998769	0.998722	0.999877	0.999872	top_n_features
decision_tree_baseline	DecisionTreeClassifier	0.998769	0.998721	0.999631	0.999616	accuracy
decision_tree_engineered_comparison	DecisionTreeClassifier	0.998769	0.998721	0.999508	0.999489	decision_tree_top1_features
decision_tree_no_engineering_comparison	DecisionTreeClassifier	0.998769	0.998721	0.999631	0.999616	decision_tree_top3_features
decision_tree_split_80_20_cv3	DecisionTreeClassifier	0.998769	0.998721	0.999631	0.999617	decision_tree_top5_features
decision_tree_split_80_20_cv5	DecisionTreeClassifier	0.998769	0.998721	0.999631	0.999616	decision_tree_top7_features
decision_tree_topall_features	DecisionTreeClassifier	0.998769	0.998721	0.999631	0.999616	decision_tree_top8_features
logreg_baseline	LogisticRegression	0.998769	0.998721	0.999877	0.999872	decision_tree_topall_features
logreg_no_engineering_comparison	LogisticRegression	0.998769	0.998721	0.999877	0.999872	logreg_top1_features
logreg_split_80_20_cv3	LogisticRegression	0.998769	0.998721	0.999754	0.999744	logreg_top3_features
logreg_split_80_20_cv5	LogisticRegression	0.998769	0.998721	0.999877	0.999872	logreg_top5_features
logreg_topall_features	LogisticRegression	0.998769	0.998721	0.999877	0.999872	logreg_top7_features
decision_tree_top5_features	DecisionTreeClassifier	0.998154	0.998081	0.999015	0.998977	logreg_top8_features
logreg_top7_features	LogisticRegression	0.998154	0.998081	0.999015	0.998977	logreg_topall_features
logreg_top8_features	LogisticRegression	0.998154	0.998081	0.999015	0.998977	random_forest_top1_features
random_forest_top5_features	RandomForestClassifier	0.996923	0.996797	0.998277	0.998208	random_forest_top3_features
decision_tree_top3_features	DecisionTreeClassifier	0.993231	0.992926	0.994584	0.994348	random_forest_top5_features
random_forest_top3_features	RandomForestClassifier	0.993231	0.992926	0.994584	0.994348	random_forest_top7_features
logreg_top3_features	LogisticRegression	0.992615	0.992278	0.994091	0.993831	random_forest_top8_features
decision_tree_top1_features	DecisionTreeClassifier	0.985846	0.985094	0.985229	0.984431	random_forest_topall_features
logreg_top1_features	LogisticRegression	0.985846	0.985094	0.985229	0.984431	
random_forest_top1_features	RandomForestClassifier	0.985846	0.985094	0.985229	0.984431	experiment_name
dummy_baseline	DummyClassifier	0.518154	0.000000	0.517971	0.000000	model_class
						test_size
						cv_folds
						accuracy
						decision_tree_split_80_20_cv3
						decision_tree_split_80_20_cv5
						decision_tree_split_70_30_cv3
						decision_tree_split_70_30_cv5
						logreg_split_80_20_cv3
						logreg_split_80_20_cv5
						logreg_split_70_30_cv3
						logreg_split_70_30_cv5
						random_forest_split_80_20_cv3
						random_forest_split_80_20_cv5
						random_forest_split_70_30_cv3
						random_forest_split_70_30_cv5

Top 10 Features by Cramér's V

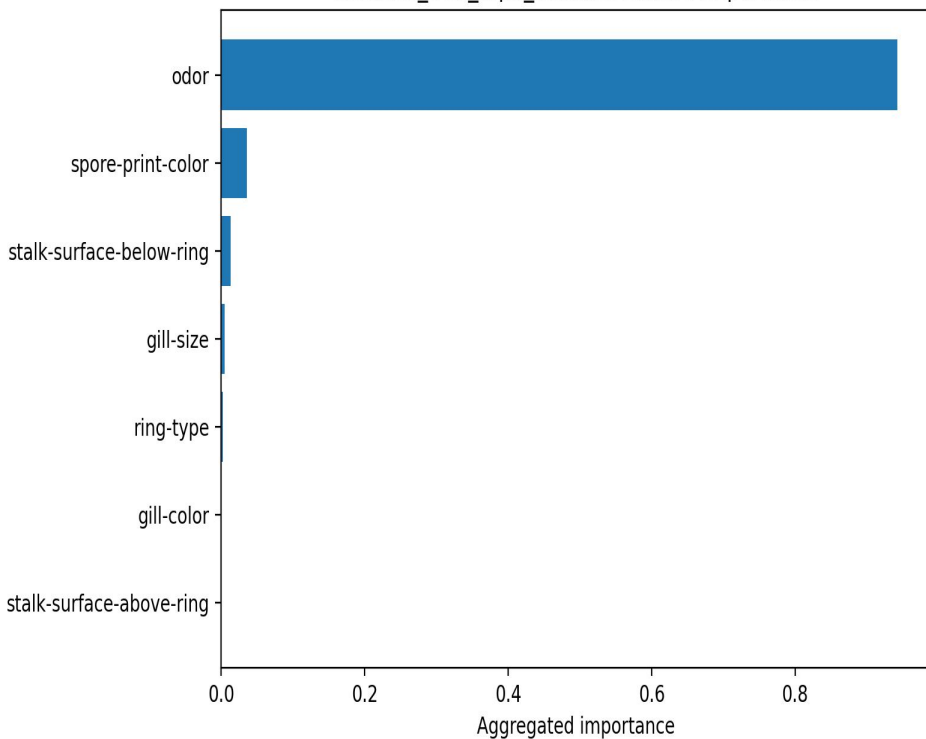


Top-k feature sweep: holdout accuracy by number of input features

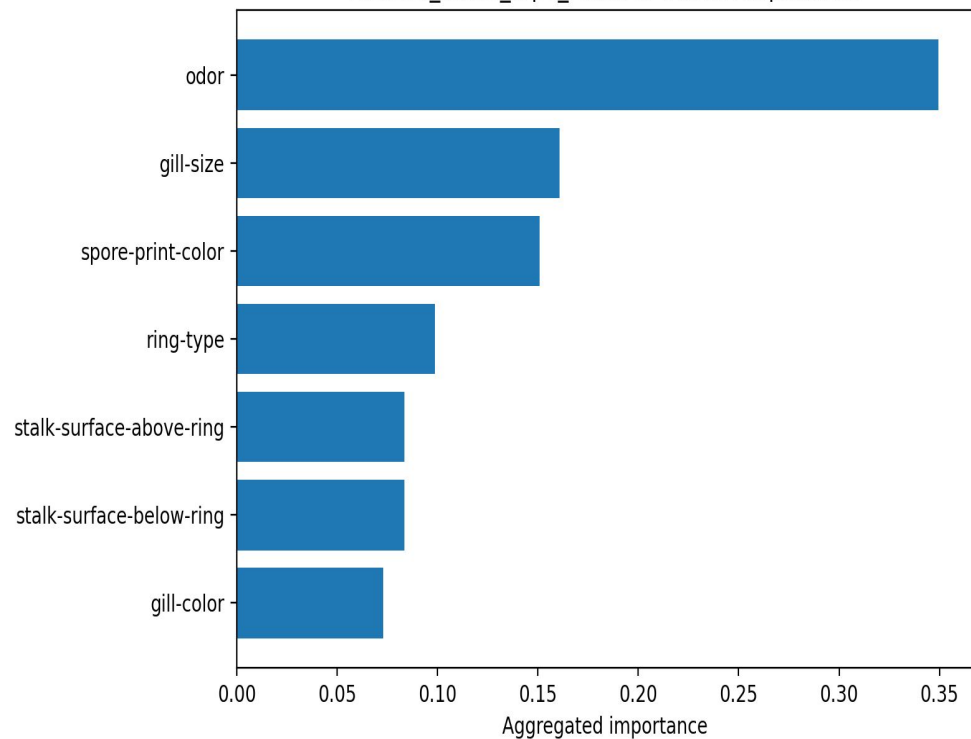


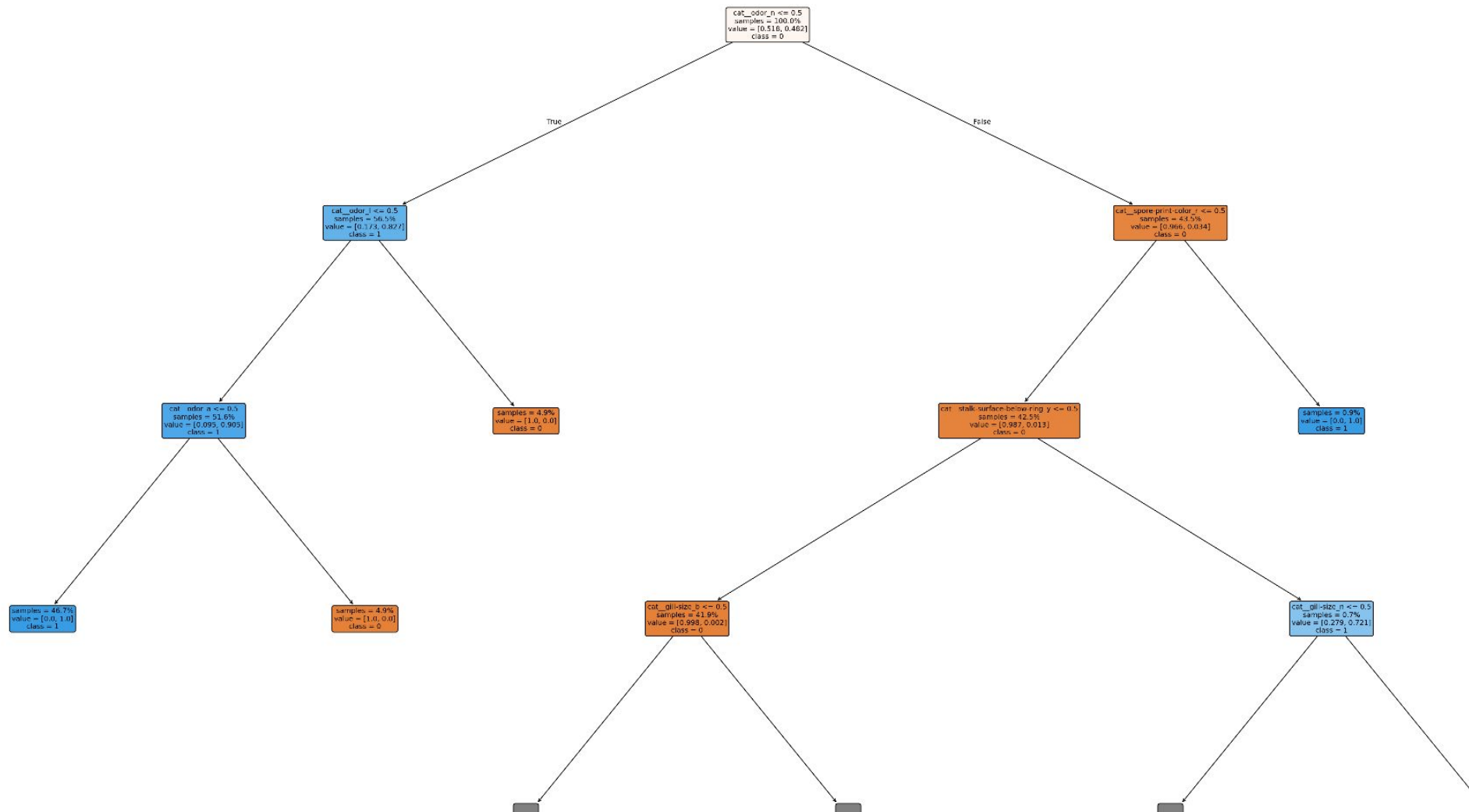


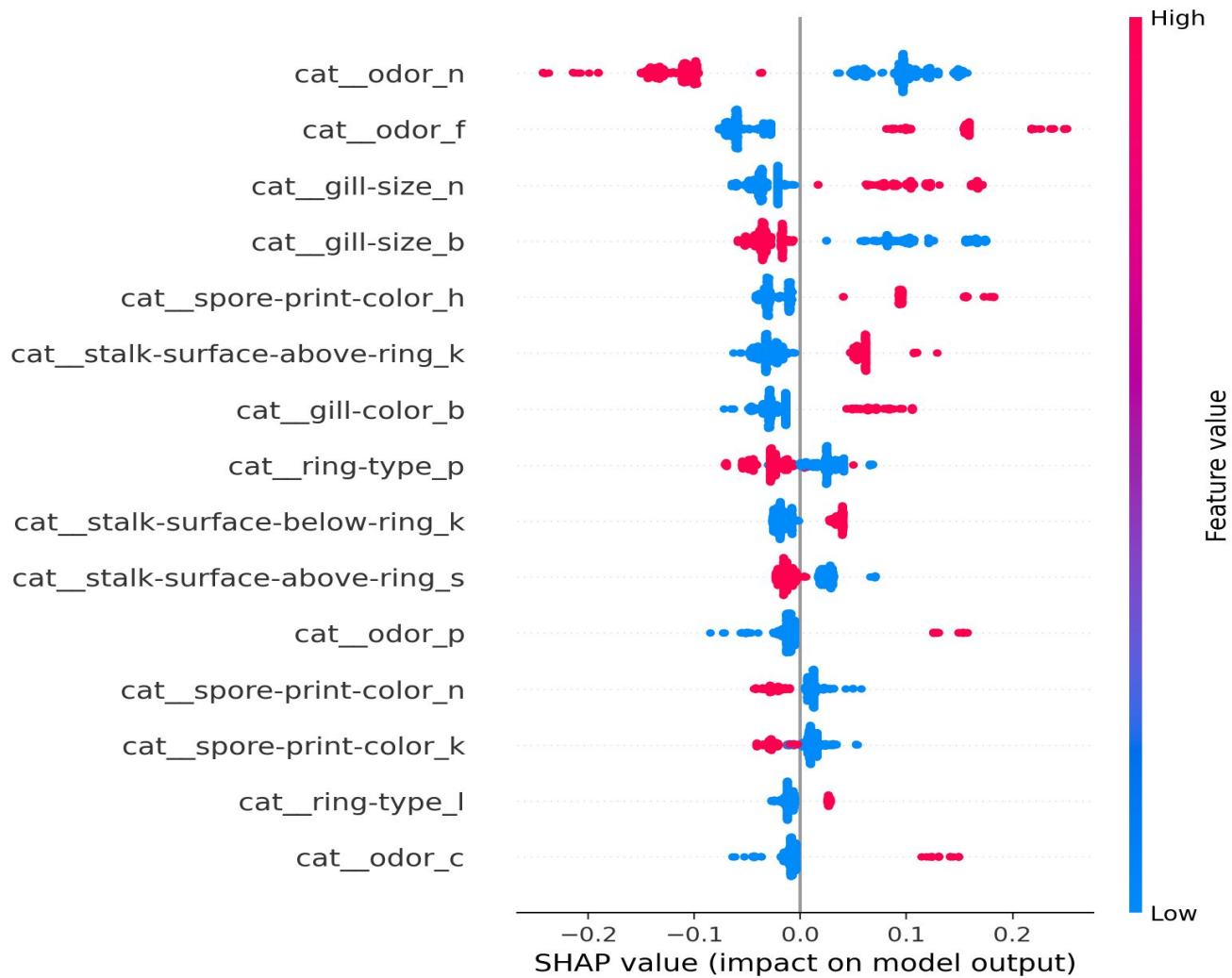
decision_tree_top7_features feature importance



random_forest_top7_features feature importance





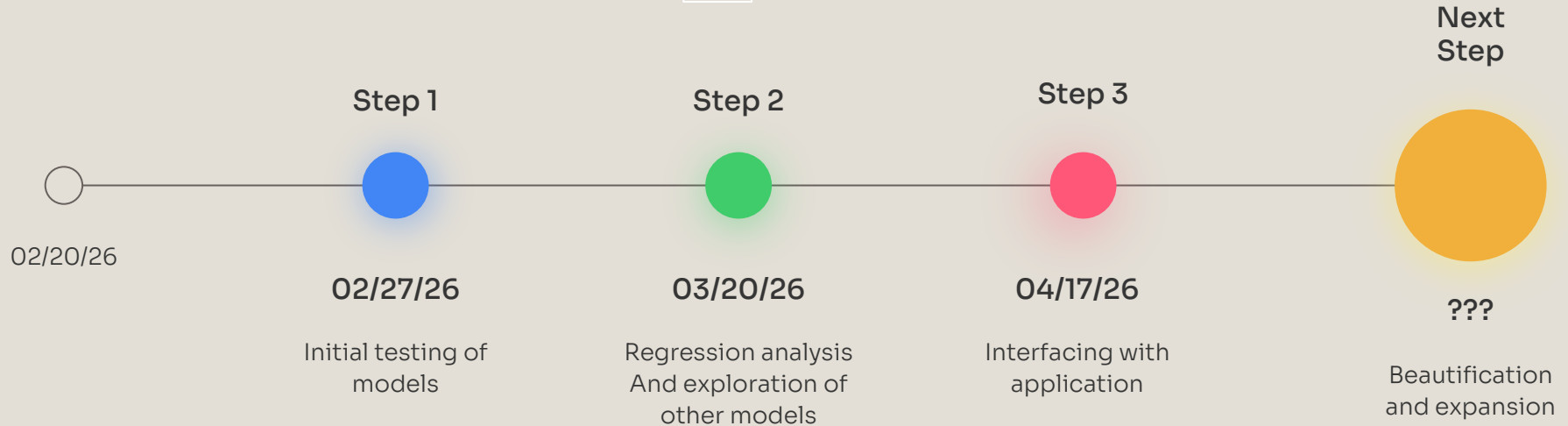


Model Deployment

Tkinter was used to build the application, so users can interact with the model without needing to use the command line.

We used mapped values to allow users to use regular words while the application converts them into the values expected by the model. To connect with the model, the interface used Pickle as well as Joblib to save and reuse the model after training was complete. This way instead of training the model every time the program starts we can train once, export to a file and load it later instantly.

Roadmap



Next steps

05-01-26

**Start
Beautification**

We're here!

Flesh out current ui look and useability of the application.

06-01-26

**Expansion to other
datasets**

Expand our model to include other datasets of mushrooms as well.

07-01-26

**Final touches and
Deployment**

Finish any final touches and reevaluate our goals to decide deployment.



Let's discuss any questions or comments

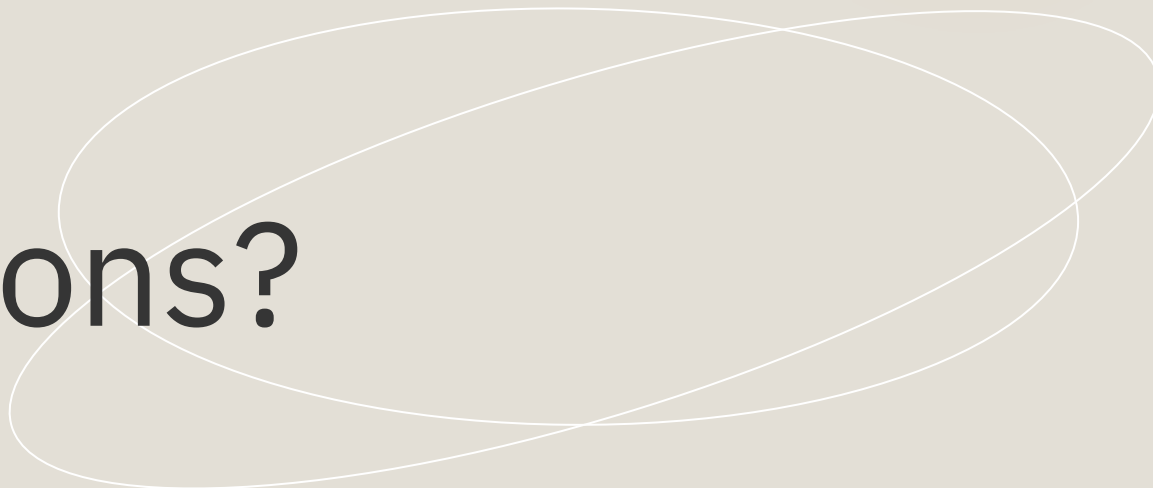
If something comes up later, contact us:

cjduckworth@mavs.coloradomesa.edu

manuttall@mavs.coloradomesa.edu

cmross@mavs.coloradomesa.edu

Any questions?
Ask away!





Thank you